Thermal Physics Formula Sheet 2025

Temperature and Zeroth Law

- $T_{\circ C} = T_K - 273.15$
- $T_{\circ F} = \frac{9}{5} T_{\circ C} + 32$
- **Linear Calibration:** $X(T) = AT + B$

Thermal Expansion

- Coefficient of Linear Expansion: $\alpha = \frac{\Delta L}{L_0 \Delta T}$
- Coefficient of Volume Expansion: $\beta = \frac{\Delta V}{V_0 \Delta T}$

Heat and Heat Capacity

- $Q = mc_m \Delta T$

Latent Heat and Phase Changes

- $Q = mL$

Heat Transfer Mechanisms

- $\frac{dQ}{dT} = \kappa A \frac{\Delta T}{L}$; $R_{th} = \frac{L}{\kappa A}$
- Thermal conductivity k:$k_{eq} = \frac{1}{2}(k_1 + k_2)$
- Thermal resistance: $\Delta T = \dot{Q} R_{Th}$ (Analogues with $V = IR$)

Thermodynamic Systems

- Number density: $\rho = \frac{N}{V}$
- Pressure equation of state: $P = f(\rho, T)$

Ideal Gases

- Boyle's Law: $PV = NK_bT = nRT$
- $K_b = 1.38 \times 10^{-23} JK^{-1}$
- $R = N_A K_b = 8.314\ J.mol^{-1}.K^{-1}$
- $n = \frac{N}{N_A} = \frac{m}{M}$
  $N_A = 6.02214076 \times 10^{23}$

Work Done in Compressing a fluid

- $F = PA$
- Work done **on the fluid** by **the system**: $dW = -PdV$
- $W_{2,1} = -\int_{V_1}^{V_2} P(T,V) dV$ (For isothermal processes)

Work Done In Ideal Gas Compression

- $W = -NK_BT\ ln \frac{V_2}{V_1} = NK_BT\ ln \frac{P_f}{P_i}$
- **For quasi-static, isothermal expansions:** $T_f = T_i$; $\frac{V_f}{V_i} = \frac{P_i}{P_f}$

First Law of Thermodynamics

- $\Delta E = W + Q$
- For isobaric processes: $dE = dQ - PdV$
- Internal energy for **ideal gasses**: $E = \frac{3}{2} NK_BT$

Heat Capacities

- Average Heat Capacity: $C_{ave} = \frac{Q}{\Delta T}$
- Isochoric heat capacity: $C_V = \left(\frac{\partial E}{\partial T}\right)_V$
- Isochoric Heat Capacity for **ideal gasses**: $C_V = \frac{3}{2} NK_B$

Specific Heats

Molar-specific isochoric heat capacity for ideal gasses:
$C_{nV} = \frac{3}{2} N_A K_B$

Enthalpy and Heat Capacity

- Enthalpy: $H = E + PV$
- $dQ = dE + PdV$
- $dH = d(E + PV) = dQ$
- Isobaric Heat Capacity: $C_p = \left(\frac{\partial H}{\partial T}\right)_P$
- $E = E(T, V)$
- $C_P = C_V + \left(\frac{\partial E}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P + P \left(\frac{\partial V}{\partial T}\right)_P$

Heat Capacities of an Ideal Gas

- $C_P = \frac{5}{2} NK_B$
- $C_V = \frac{3}{2} NK_B$

Adiabatic Processes in an Ideal Gas

- $dE = dW = C_V dT$
- $N_A K_B = R = C_P - C_V$
- $\gamma = \frac{C_P}{C_V}$
- $T_f V_f^{\gamma - 1} = T_0 V_0^{\gamma - 1} = constant$
- $PV^\gamma = P_0 V_0^\gamma = constant$
- $TP^{\frac{1-\gamma}{\gamma}} = T_0 P_0^{\frac{1-\gamma}{\gamma}} = constant$
- For mono-atomic ideal gasses: $\gamma = \frac{5}{3}$
- $W = -\frac{P_2V_2 - P_1V_1}{1-\gamma} = -\frac{nR(T_2 - T_1)}{1-\gamma}$
- $\left(\frac{\partial P}{\partial V}\right)_{\Delta Q = 0} = -\gamma \frac{P}{V}$; $\left(\frac{\partial P}{\partial V}\right)_T = -\frac{P}{V}$

The Second Law of Thermodynamics

- For isothermal expansion: $\Delta E = 0$
- $S_{composite} = S_A + S_B = S(N_A, E_A, V_A, N_B, E_B, V_B)$
- $\Delta S_{composite} \geq 0$

Thermodynamic temperature

- $\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{V,N}$
- $\Delta S = \left[\frac{1}{T_A} - \frac{1}{T_B}\right] \Delta E_A > 0$

Heat Engines and the 2nd Law

- $dS = \left(\frac{\partial S}{\partial E}\right)_{V,N} dE$
- For pure heating: $dE = dQ$
- $dS = \frac{dQ}{T}$; $S = \frac{Q}{T}$
- $W = Q_{high} - Q_{low}$
- $\Delta S_{total} = \Delta S_{high} + \Delta S_{low} = -\frac{Q_{high}}{T_{high}} + \frac{Q_{low}}{T_{low}} \geq 0$
- $\frac{Q_{low}}{Q_{high}} \geq \frac{T_{low}}{T_{high}}$
- Heat efficiency $\eta = \frac{W}{Q_{high}} = 1 - \frac{Q_{low}}{Q_{high}}$

Carnot Cycle of an Ideal Gas

- $\eta_{max} = 1 - \frac{T_{low}}{T_{high}}$ $(\Delta S = 0)$
- Isothermal Expansion: $Q = -W_{AB} = NK_BTln \frac{V_B}{V_A}$
- $W_{BC} = \frac{3}{2} NK_B (T_C - T_B)$
- $T_{high} V_B^{\gamma - 1} = T_{low} V_C^{\gamma - 1}$
- $\frac{T_{high}}{T_{low}} \implies \frac{V_B}{V_A} = \frac{V_C}{V_D}$

Entropy Changes

- $dS = \frac{dQ}{T}$
- $dQ = C_V dT$
- $dS = C_V \frac{dT}{T}$
- $\Delta S_{21} = S_2 - S_1 = C_V \ln \frac{T_2}{T_1}$

Entropy Change in Free Expansion

- $Q_{BA} = NK_BT \ln \frac{V_B}{V_A}$
- $\Delta S_{BA} = NK_B \ln \frac{V_B}{V_A}$

Thermodynamic and Gas Temperatures

- Real Gas temperature: $\theta = \frac{PV}{NK_B}$
- $dQ = dE - PdV$
- $\frac{dQ}{\theta} = \frac{dE}{\theta} - \frac{NK_B dV}{V}$

The Carnot cycle is a closed cycle

- $\oint \frac{dQ}{\theta} = \frac{Q_{cold}}{\theta_{cold}} - \frac{Q_{hot}}{\theta_{hot}} = 0$
- $\frac{\theta_{hot}}{\theta_{cold}} = \frac{T_{hot}}{T_{cold}}$

Thermodynamic Pressure

- In thermal equilibrium, to maximise entropy:
  $dS_{total} = \left(\frac{\partial S_A}{\partial V_A}\right)_{E_A N_A} dV_A + \left(\frac{\partial S_B}{\partial V_B}\right)_{E_B N_B} dV_B = 0$
- $dV_A = -dV_B$
- $\frac{P}{T} = \left(\frac{\partial S}{\partial V}\right)_{E,N}$
- Chemical Potential: $\frac{\mu}{T} = -\left(\frac{\partial S}{\partial N}\right)_{E,V}$

Fundamental Thermodynamic Relation

- $dS = \left(\frac{\partial S}{\partial E}\right)_{V,N} dE + \left(\frac{\partial S}{\partial V}\right)_{E,N} dV + \left(\frac{\partial S}{\partial N}\right)_{E,V} dN$
- $dS = \frac{1}{T} dE + \frac{P}{T} dV - \frac{\mu}{T} dN$
- $dE = TdS - PdV + \mu dN$
- Intensive variables:
- $T = \left(\frac{\partial E}{\partial S}\right)_{V,N}$; $P = -\left(\frac{\partial E}{\partial V}\right)_{S,N}$; $\mu = \left(\frac{\partial E}{\partial N}\right)_{S,V}$
- $dS = \left(\frac{\partial E}{\partial T}\right)_{V,N}$ $(dV = 0)$

Entropy of an Ideal Gas

- $\frac{1}{T} = \frac{3NK_B}{2E}$
- $\frac{P}{T} = \frac{NK_B}{V}$
- $dS = \frac{3}{2} Nk_B \frac{dE}{E} + Nk_B \frac{dV}{V}$
- $\Delta S = \frac{3}{2} Nk_B \ln(\frac{E_2}{E_1}) + Nk_B \ln(\frac{V_2}{V_1})$

Third Law of Thermodynamics

- $\lim_{T \to 0} S = 0$
- $S(T_2, V) - S(T1, V) = \int_{T_1}^{T_2} \frac{C_V(T)}{T} dT$

Helmholtz Free Energy (F)

- $\Delta S_{bath} = -\frac{Q}{T_{bath}}$
- $\Delta S_{total} = \Delta S - \frac{Q}{T_{bath}} \geq 0$
- Availability: $A = E + P_{bath}V - T_{bath}S$
- $\Delta A = \Delta E - T\Delta S \leq 0$
- $F = E - TS$
- $\Delta F \leq 0$
- $dF = -SdT - PdV + \mu dN$
- $S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}$, $P = -\left(\frac{\partial F}{\partial V}\right)_{T,N}$

Gibbs Free Energy (G)

- At mechanical equilibrium: $P = P_{bath}$
- At mechanical equilibrium: $\Delta A \equiv \Delta G$
- $G = E + PV - TS = F + PV$
- $v = \frac{V}{N}$, $s = \frac{S}{N}$
- $dg = (\frac{\partial g}{\partial P})_T dP + (\frac{\partial g}{\partial T})_P dT = vdP - sdT$
- $dG = -SdT + VdP + \mu dN$

Enthalpy (H)

- $H = E + PV$
- $dH = TdS + VdP + \mu dN$
- $T = \left(\frac{\partial H}{\partial S}\right)_{P,N}$, $V = \left(\frac{\partial H}{\partial P}\right)_{S,N}$, $\mu = \left(\frac{\partial H}{\partial N}\right)_{S,V}$

Landau Potential (Ω)

- $\Omega(T, V, \mu) = E - TS - \mu N = F - \mu N$
- $d\Omega = -SdT - PdV - Nd\mu$
- $-S = \left(\frac{\partial \Omega}{\partial T}\right)_{V,\mu}$, $P = -\left(\frac{\partial \Omega}{\partial V}\right)_{S,\mu}$, $N = -\left(\frac{\partial \Omega}{\partial \mu}\right)_{T,V}$

Useful Work and Availability

- $W_{useful} \leq -(\Delta E + P_{bath}\Delta V - T_{bath}\Delta S) = -\Delta A$
- $W = P_{bath}\Delta V + W_{useful}$

Maxwell Relations

- $\frac{\partial}{\partial S}\left[\frac{\partial E}{\partial V}\right] = \frac{\partial}{\partial V}\left[\frac{\partial E}{\partial S}\right]$
- First: $\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial S}\right)_V$
- Second: $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$
- Third: $\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$
- Fourth: $\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$
- Fifth: $\left(\frac{\partial V}{\partial N}\right)_P = \left(\frac{\partial \mu}{\partial P}\right)_N$
- Sixth: $\left(\frac{\partial \mu}{\partial V}\right)_N = -\left(\frac{\partial P}{\partial N}\right)_V$

Applying the Maxwell relations

- $K_T = -\frac{1}{V}(\frac{\partial V}{\partial P})_T$
- $\alpha = \frac{1}{V}(\frac{\partial V}{\partial T})_P$
- $C_P - C_V = \frac{VT\alpha^2}{K_T}$
- $C_V = (\frac{\partial E}{\partial T})_V = T(\frac{\partial S}{\partial T})_V$
- $C_P = (\frac{\partial H}{\partial T})_P = T(\frac{\partial S}{\partial T})_P$
- $\left(\frac{\partial E}{\partial V}\right)_T = T\left(\frac{\partial E}{\partial T}\right)_V - P$

Van der Waals Gases

- Lennard-Jones potential: $U(r) \propto \frac{\sigma^{12}}{r^{12}} - \frac{\sigma^6}{r^6}$
- Effective Pressure: $P_{eff} = P + \frac{N^2}{V^2} a$
- $V_{eff} = V - Nb$
- $P_{eff} V_{eff} = Nk_bT$
- $\left(P + \frac{N^2}{V^2} a\right)(V - Nb) = Nk_bT$
- $T\left(\frac{\partial P}{\partial T}\right)_V - P = \frac{N^2 a}{V^2}$
- $H_2O:$ $a = 0.551\ J.m^3.mol^{-2}$; $b = 3.04 \times 10^{-6}\ m^3.mol^{-1}$
- $He:$ $a = 0.003\ J.m^3.mol^{-2}$; $b = 2.34 \times 10^{-6}\ m^3.mol^{-1}$

Internal Energy of Van der Waals Gas

- PES for VDW gases: $P = \frac{Nk_bT}{V - Nb} - \frac{aN^2}{V^2}$
- $\left(\frac{\partial E}{\partial V}\right)_T = \frac{N^2 a}{V^2}$
- $E_{vdw} = E_{ideal} - \frac{N^2 a}{V}$

Critical Temperature, Pressure and Volume

- $(\frac{\partial P}{\partial V})_T = 0, (\frac{\partial^2 P}{\partial V^2})_T = 0$
- $V_c = 3Nb$
- $T_C = \frac{8a}{27k_B b}$ and $P_C = \frac{a}{27b^2}$

Free Expansion of Gases

- $dE = (\frac{\partial E}{\partial T})_V dT + (\frac{\partial E}{\partial V})_T = 0$
- Cyclic Relation: $\left(\frac{\partial T}{\partial V}\right)_E \left(\frac{\partial E}{\partial T}\right)_V = -\left(\frac{\partial E}{\partial V}\right)_T$
- $\left(\frac{\partial T}{\partial V}\right)_E = -\frac{1}{C_V}[T(\frac{\partial P}{\partial T})_V - P] \implies \Delta T_{21} = \int_{V_1}^{V_2} \left(\frac{\partial T}{\partial V}\right)_E dV$
- $E = \frac{3}{2} Nk_BT - \frac{N^2 a}{V^2}$
- $\Delta T = -\int_{V_1}^{V_2} \frac{2N_A}{3V^2 K_B} dV < 0$

Joule-Kelvin Expansion

- $W = \Delta E = -\int_{V_1}^{0} PdV - \int_0^{V_2} PdV = P_1V_1 - P_2V_2$
- $H_1 = H_2$
- $d[H(T, P)] = 0$
- $\left(\frac{\partial T}{\partial P}\right)_H = -\left(\frac{\partial H}{\partial P}\right)_T \left(\frac{\partial T}{\partial H}\right)_P$
- $C_P = \left(\frac{\partial H}{\partial T}\right)_P \implies \left(\frac{\partial T}{\partial H}\right)_P = \frac{1}{C_P}$
- $\left(\frac{\partial H}{\partial P}\right)_T = T\left(\frac{\partial S}{\partial P}\right)_T + V$
- $\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P$
- $\left(\frac{\partial T}{\partial P}\right)_H = \frac{V(T\alpha - 1)}{C_p}$
- $\Delta T = \int_{P_1}^{P_2} \frac{V}{C_p}(T\alpha - 1) dP$
- $T < \frac{2a}{bk_b} \implies$ cooling
- $T > \frac{2a}{bk_b} \implies$ heating
- **For ideal gases:** $\alpha = \frac{1}{T} \implies \Delta T = 0$
- **For VDW gases:** $\rho = \frac{N}{V}$
- $P + \alpha\rho^2 = \frac{\rho K_B T}{1 - b\rho}$
- $\alpha = -\frac{1}{\rho}\left(\frac{\partial \rho}{\partial T}\right)_P$
- $\alpha = \frac{1}{T}(1 - b\rho)\left(1 - \frac{2\alpha\rho(1 - b\rho)^2}{k_BT}\right)^{-1}$
- **For low number densities:** $\alpha \approx \frac{1}{T}\left[1 - \rho\left(b - \frac{2a}{k_bT}\right)\right]$

Phase Equilibrium

- **Thermal Equilibrium:** $T_1 = T_2$
- **Mechanical Equilibrium:** $P_1 = P_2$
- **Chemical Equilibrium:** $\mu_1 = \mu_2$
- $\mu(T, P) = g(T, P)$
- **Along phase boundaries:** $g_1(T, P) = g_2(T, P)$

Claussius-Clapeyron Equation

- $\Delta g_1 = \Delta g_2$
- $dg = -sDT + \nu dP \implies \Delta g = -s\Delta T + \nu \Delta P$
- **CCE:** $\frac{dP}{dT} = \frac{s_2 - s_1}{\nu_2 - \nu_2} = \frac{\Delta S}{\Delta V} = \frac{\Delta h_{fusion}}{T \Delta V}$

Enthalpy & Latent Heat

- $T\frac{\partial S}{\partial V} = \frac{\partial E}{\partial V} + P$
- $T\Delta S = \Delta E + P\Delta V$
- $L_{21} = \Delta H = T\Delta S = \Delta E + P\Delta V$
- $L_{fusion} = T(S_{liquid} - S_{solid})$
- **Discontinuity in S $\implies$ phase transition**
- $\frac{dP}{dT} = \frac{L}{T\Delta V} = \frac{l^{(\text{molar latent heat})}}{T\Delta\nu^{(\text{specific volume})}}$

Vapour Pressure Curve

- $\Delta V = V_{gas} - V_{liquid} \approx V_{gas}$; **For ideal gases:** $V_{gas} \approx \frac{RT}{P}$
- $P(T) = P_0 \exp[-\frac{1}{RT}]$

Math

$\left(\frac{\partial x}{\partial y}\right)_z \times \left(\frac{\partial y}{\partial z}\right)_x \times \left(\frac{\partial z}{\partial x}\right)_y = -1$; $(\frac{\partial y}{\partial x})_z = \frac{1}{(\frac{\partial x}{\partial y})_z}$

$\int \sin x\, dx = -\cos x + C$; $\int \cos x\, dx = \sin x + C$;
$\int \tan x\, dx = -\ln|\cos x| + C$; $\int \cot x\, dx = \ln|\sin x| + C$;
$\int \sec x\, dx = \ln|\sec x + \tan x| + C$; $\int \csc x\, dx = \ln|\csc x - \cot x| + C$
$\int \sec^2 x\, dx = \tan x + C$; $\int \csc^2 x\, dx = -\cot x + C$;
$\int \sec x \tan x\, dx = \sec x + C$; $\int \csc x \cot x\, dx = -\csc x + C$
$\int \sinh x\, dx = \cosh x + C$; $\int \cosh x\, dx = \sinh x + C$;
$\int \tanh x\, dx = \ln|\cosh x| + C$

$\frac{d}{dx} sinx = cosx$; $\frac{d}{dx} cosx = -sinx$; $\frac{d}{dx} tanx = sec^2 x$;
$\frac{d}{dx} cotx = -csc^2 x$; $\frac{d}{dx} secx = sec(x)tan(x)$
$\frac{d}{dx} cscx = -csc(x)cot(x)$; $\frac{d}{dx} sinhx = coshx$; $\frac{d}{dx} coshx = sinhx$;
$\frac{d}{dx} tanhx = sech^2 x$; $\frac{d}{dx} cothx = -csch^2(x)$;
$\frac{d}{dx} sechx = -sech(x)tanh(x)$; $\frac{d}{dx} cschx = -csch(x)coth(x)$

$\sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$
$\cosh(x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$
$\coth^2 x - 1 = csch^2 x$; $1 - \tanh^2 x = sech^2 x$; $\cosh^2 x - \sinh^2 x = 1$

Constants and Conversions

$R = 8.314\,462\,618\ \mathrm{J\,mol^{-1}\,K^{-1}}$; $k_B = 1.380\,649 \times 10^{-23}\ \mathrm{J\,K^{-1}}$
$N_A = 6.022\,140\,76 \times 10^{23}\ \mathrm{mol^{-1}}$; $R = N_A \cdot k_B$
$c = 4.18\ \mathrm{J\,g^{-1}\,K^{-1}}$ or $4180\ \mathrm{J\,kg^{-1}\,K^{-1}}$
$1\ \mathrm{atm} = 101\,325\ \mathrm{Pa}$; $1\ \mathrm{l} = 1 \times 10^{-3}\ \mathrm{m^3}$
$1\ \mathrm{mL} = 1 \times 10^{-6}\ \mathrm{m^3}$; $1\ \mathrm{bar} = 1 \times 10^5\ \mathrm{Pa}$
$1\ \mathrm{cm^3} = 1 \times 10^{-6}\ \mathrm{m^3}$; $1\ \mathrm{mL} = 1\ \mathrm{cm^3}$
$1\ \mathrm{eV} = 1.602\,176\,634 \times 10^{-19}\ \mathrm{J}$; $1\ \mathrm{V} = 1\ \mathrm{J\,C^{-1}}$